

SRI SHANMUGHA COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Content Beyond Syllabus
Academic Year (2025 – 26)

Name of the Faculty	: R.ASHA
Subject Code / Subject Name	: 23AD410 / Data Exploration and Visualization
Year / Semester	: II / III
Batch	: 2024 – 2028

UNIT I – EXPLORATORY DATA ANALYSIS

Beyond-the-Syllabus Topics:

1. EDA in Healthcare and Finance – Real-world applications of EDA in medical diagnostics, fraud detection, and stock market analysis.
2. Automated EDA Tools – Overview of tools like Pandas Profiling, Sweetviz, and D-Tale for rapid exploratory analysis.
3. Data Quality Assessment – Techniques to detect and handle missing values, duplicates, and inconsistent data in large datasets.
4. EDA in Big Data Environments – Applying EDA concepts using PySpark and Dask for distributed datasets.
5. Feature Interaction Discovery – Identifying hidden patterns between features using correlation matrices and heatmaps.

UNIT II – VISUALIZING USING MATPLOTLIB

Beyond-the-Syllabus Topics:

6. Interactive Visualizations with Plotly – Creating dynamic, web-ready charts that go beyond static Matplotlib plots.
7. Dashboarding with Dash and Streamlit – Building interactive data dashboards for real-time data exploration.
8. Advanced Seaborn Techniques – Pair plots, cluster maps, and FacetGrid for multi-dimensional visual analysis.
9. Color Theory in Data Visualization – Principles of perceptual color design and accessibility (colorblind-friendly palettes).
10. Animation in Matplotlib – Creating animated plots using FuncAnimation to visualize changes over time.

UNIT III – UNIVARIATE ANALYSIS

Beyond-the-Syllabus Topics:

11. Kernel Density Estimation (KDE) – Non-parametric method to estimate probability density functions of a single variable.
12. Outlier Detection Techniques – Using Z-score, IQR, and isolation forest for identifying anomalies in univariate data.
13. Statistical Hypothesis Testing – Applying t-tests, chi-square, and Shapiro-Wilk tests to validate distribution assumptions.
14. Lorenz Curve and Gini Coefficient – Measuring inequality in income and wealth distribution using Python.
15. Smoothing Techniques Beyond Moving Averages – LOWESS and Savitzky-Golay filters for noise reduction in time series.

UNIT IV – BIVARIATE ANALYSIS

Beyond-the-Syllabus Topics:

16. Partial Correlation Analysis – Understanding relationships between two variables while controlling for a third.
17. Simpson’s Paradox – How aggregated data trends can reverse when data is split into subgroups – real-world examples.
18. Scatter Plot Matrices (SPLOM) – Visualizing pairwise relationships across multiple numeric variables simultaneously.
19. Robust Regression Techniques – Handling outliers in bivariate data using Theil-Sen and Huber regression.
20. Association Rule Mining Basics – Discovering relationships between categorical variables using support and confidence metrics.

UNIT V – MULTIVARIATE AND TIME SERIES ANALYSIS

Beyond-the-Syllabus Topics:

21. Dimensionality Reduction Techniques – PCA, t-SNE, and UMAP for visualizing high-dimensional multivariate data.
22. Anomaly Detection in Time Series – Using ARIMA residuals and isolation forest to detect unusual patterns.
23. Geospatial Multivariate Analysis – Combining geographic and demographic data for location-based insights using Folium.
24. Time Series Forecasting – Introduction to ARIMA, SARIMA, and Facebook Prophet for future trend prediction.
25. EDA in Clinical Trials and Epidemiology – Applying multivariate analysis to healthcare datasets for patient outcome analysis.

COURSE INSTRUCTOR	HOD
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